Vamshi Narayana Babu

(623) 299-0980 | vamshin24@asu.edu | linkedin.com/in/vamshin24 | github.com/vamshin24 | Web portfolio

EDUCATION

Arizona State University, Tempe, AZ — May 2026
Master of Science in Robotics and Autonomous Systems | GPA: 3.72/4
Coursework: Python, Machine Learning, Kinematics and Dynamics, Advanced Data Structures and Algorithms

National Institute of Technology Karnataka, Surathkal, India — Jul 2018-May 2022
Bachelor of Technology in Electronics and Communication Engineering

TECHNICAL SKILLS

Languages: Python, Java, MATLAB, SQL, JavaScript, TypeScript, Bash
Robotics and Automations: ROS, ROS2, Gazebo, Mujoco, MoveIt, RViz, Nav2, SLAM, Cartographer, PID Controller, HIL Simulation, Aruco markers, Sensor Integration (LIDAR, IMU, Camera), Pydobot
Computer Vision and ML: OpenCV, PyTorch, TensorFlow, SciKit-Learn, Keras, Huggingface Transformers, FashionCLIP, Pytesseract (OCR), Redis, Kafka, RabbitMQ, Django, SpringBoot, REST APIs, Hibernate
Tools and Frameworks: Docker, Kubernetes, Git, GitHub, Jenkins, CI/CD Pipelines, Multi-threading, Postman, DBeaver
Cloud and Databases: AWS, Azure, Docker, Kubernetes, MongoDB, PostgreSQL, MySQL
Frontend and UI Tools: React, Next.js, Tailwind CSS, Streamlit, Framer Motion

WORK EXPERIENCE

Stealth Startup, San Francisco, CA, USA — Jun 2025-Aug 2025
Robotics Software Engineer Intern
- Built a real-time mobile navigation system for the **robotic MVP** using **ROS2 Nav2**, **LiDAR SLAM**, and **Python**, achieving 90% aisle coverage and 98% picking success during live grocery store testing.
- Enabled intelligent pick-and-place across **40+ SKUs** by building a dual-camera vision pipeline and training a **VLA model** on 300+ demos, improving grasp accuracy by **25%** and picking efficiency by **41%** using **OpenCV**.
- Improved learning efficiency by **38%** over imitation-only pipelines by implementing **HIL-SERL** on the SO101 arm using **reinforcement learning** with real-time visual feedback loops.
- Integrated vision, navigation, and manipulation modules using **ROS2**, **Python**, and **custom APIs**, reducing overall system integration time by **30%** ahead of investor demonstration deadlines.

Addverb Technologies, Noida, UP, India — Aug 2022-Jul 2024
Robotics Software Engineer
- Built warehouse automation solution integrating AMRs, ASRS, WCS in **Java, Spring Boot, MSSQL**, enhancing inventory tracking, task scheduling and robot fleet coordination while efficiently processing 10K+ orders daily.
- Optimized messaging with **Redis, Kafka, RabbitMQ**, improving coordination by **15%** between **Zippy** robots.
- Developed a backend-centric DWS system using **Spring Boot** and **REST APIs**, integrating **TCP/IP** protocols with conveyor PLCs to improve data handling, achieving **80%** workflow automation and **20%** latency reduction.
- Enhanced warehouse storage efficiency by **35%** by refactoring WCS scheduling logic in **Python** and **Django**, resolving 70+ bugs and improving system reliability.
- Led the architecture for an inbound palletization system using Spring Boot and Maven, boosting throughput and warehouse efficiency by 40%.
- Transformed **15,000 images per day** to aid an automotive in selling warehouse robots, by deploying an OpenCV workflow with Flask APIs and AWS SageMaker using U2Net and ARShadowGAN models.

PROJECTS

Autonomous SLAM and Teleoperation with Turtlebot3 and ROS2
- Engineered a real-time **SLAM** and teleoperation system for TurtleBot3 using **ROS2**, developing a custom **Python GUI** and integrating **Cartographer** for autonomous mapping in **Gazebo** and real-world environments, enabling robust navigation and task-level autonomy.

UAV Autonomous Landing on moving platform and Color-Based Navigation
- Designed Motor mixing Algorithm utilizing **OpenCV, Python, MATLAB** for motion control, increasing stability, adaptive control, precision and dynamic path optimization achieving **97%** landing accuracy in 50+ test flights.

Path planning for Maze Solving with MyCobot Pro 600 Robotic Arm
- Programmed a robot for autonomous navigation, motion planning and path optimization (**A***), incorporating inverse kinematics with **Python, OpenCV, MATLAB Simulink, ROS2**, reducing maze-solving time by **40%**.

Swipe-to-Shop Progressive Web App & ML Recommendation Engine | React, PyTorch
- Delivered personalized fashion recommendations with ~22% accuracy improvement and sub-200ms latency by fine-tuning **FashionCLIP**, normalizing product data, and implementing randomized **KNN** for adaptive swipes.

LEADERSHIP ROLES

- **Runner-up, Devils Invent Hackathon** - Sponsored by Los Alamos National Laboratory (LANL) - Automated MiniFlow Ion Cell unpacking system with **Dobot Magician Lite, AI, Python** and **pydobot** for more efficiency.
- **Visual AI Hackathon Winner**(Feb2025)- OCR-based AI model for food safety using **Python**, **OCR, Pytesseract**.